INCH-POUND
MIL-W-8982/2A
27 February 1989
SUPERSEDING
MIL-W-8982/2
28 August 1986

MILITARY SPECIFICATION SHEET

WRENCH, BOX, SPLINED, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the box wrenches described herein shall consist of this document and the latest issue of MIL-W-8982.

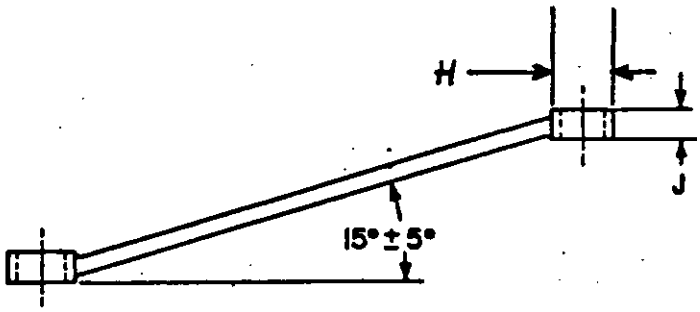


Figure 1. Box wrench, double head, 15-degree offset, each end.

DESIGN REQUIREMENTS:

1. Box wrench dimensions shall conform to Figure 1, Table I and the applicable portions of GGG-W-636. Spline and chamfer dimensions shall conform to MS33787. This wrenching design fits fasteners with the corresponding MS33787 spline.

2. The material of the box wrench shall be Grade E4340 (UNS G43406) per AMS 6415; Grade 4140 (UNS G41400) per AMS 6381; or Grade 6150 (UNS G61500) per AMS 6448.

3. The box wrench shall be hardened to 48-54 HRC.

4. The box wrench shall be chromium plated in accordance with QQ-C-320, Class 1, Type I. Cadmium plating is not permitted. Nickel underplating in accordance with QQ-N-290, Class 1, Grade G.

5. Surface roughness shall be as specified in GGG-W-636.

PERFORMANCE REQUIREMENTS:

1. The box wrench shall withstand the endurance torque load and ultimate torque load values specified in table I. There shall be no damage or permanent deformation (set) to the wrench.

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2. When engaged with spline mandrels, the box wrench shall be tested for conformance to the endurance torque load and ultimate torque load values specified in table I. The load shall be held for 10 seconds. Mandrel dimensions in the engaged portion shall be in accordance with MS33787, Table I, except that C and W diameters shall be held to a minimum dimension (+0.002, -0.000 inches).

3. Mandrels shall be hardened to not less than 55 HRC and smoothly finished in engagement area of wrench openings.

4. The box wrench shall be subjected to not less than 500 torque cycles on the spline mandrels in accordance with the endurance torque load and ultimate torque load values specified in Table I. Engagement lengths shall be the same as that specified for the ultimate tests.

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TABLE I. Box wrench dimensions and box end test loads.

Coordination number	Basic WR size		Outside diameter of head (max)		Thickness of heads (max)		Spline test							
							Overall length (in.)		Small Head			Large Head		
	Small head (in.)	Large head (in.)	H Small head (in.)	H Large head (in.)	J Small head (in.)	J Large head (in.)	Min.	Max	Ultimate torque (in.-lbs)	Endurance torque (in.-lbs)	L Engage length (in.)	Ultimate torque (in.-lbs)	Endurance torque (in.-lbs)	L Engage length (in.)
01	0.218	0.281	0.359	0.447	0.210	0.235	4.000	5.000	235	140	0.110	365	220	0.140
01A	0.218	0.281	0.359	0.447	0.210	0.235	5.550	7.500	235	140	0.110	365	220	0.140
02	0.250	0.312	0.400	0.490	0.240	0.250	4.000	5.000	300	195	0.125	450	250	0.156
02A	0.250	0.312	0.400	0.490	0.240	0.250	6.500	8.125	300	195	0.125	450	250	0.156
03	0.312	0.375	0.490	0.590	0.250	0.343	4.000	5.000	450	250	0.156	735	515	0.188
03A	0.312	0.375	0.490	0.590	0.250	0.343	6.500	8.875	450	250	0.156	735	515	0.188
04	0.375	0.438	0.590	0.680	0.343	0.375	4.185	5.185	735	515	0.188	1,250	875	0.219
04A	0.375	0.438	0.590	0.680	0.343	0.375	6.500	9.875	735	515	0.188	1,250	875	0.219
05	0.438	0.500	0.680	0.745	0.375	0.375	4.250	5.250	1,250	875	0.219	2,250	1,575	0.250
05A	0.438	0.500	0.680	0.745	0.375	0.375	7.500	10.625	1,250	875	0.219	2,250	1,575	0.250
06	0.438	0.562	0.680	0.825	0.375	0.406	4.500	5.500	1,250	875	0.219	3,000	2,275	0.281
06A	0.438	0.562	0.680	0.825	0.375	0.406	7.500	11.000	1,250	875	0.219	3,000	2,275	0.281
07	0.500	0.562	0.745	0.825	0.375	0.406	4.625	5.625	2,250	1,575	0.250	3,000	2,275	0.281
07A	0.500	0.562	0.745	0.825	0.375	0.406	7.500	11.500	2,250	1,575	0.250	3,000	2,275	0.281
08	0.562	0.625	0.825	0.907	0.406	0.468	5.000	6.000	3,000	2,275	0.281	3,250	2,450	0.312
08A	0.562	0.625	0.825	0.907	0.406	0.468	8.250	12.500	3,000	2,275	0.281	3,250	2,450	0.312
09	0.625	0.688	0.907	0.996	0.468	0.516	5.250	6.250	3,250	2,450	0.312	4,250	2,975	0.344
09A	0.625	0.688	0.907	0.996	0.468	0.516	9.438	13.250	3,250	2,450	0.312	4,250	2,975	0.344
10	0.625	0.750	0.907	1.090	0.468	0.531	5.250	6.250	3,250	2,450	0.312	4,500	3,500	0.375
10A	0.625	0.750	0.907	1.090	0.468	0.531	9.438	13.750	3,250	2,450	0.312	4,500	3,500	0.375
11	0.688	0.750	0.996	1.090	0.516	0.531	10.250	11.750	4,250	2,975	0.344	4,500	3,500	0.375
11A	0.688	0.750	0.996	1.090	0.516	0.531	13.500	14.375	4,250	2,975	0.344	4,500	3,500	0.375
12	0.750	0.812	1.090	1.170	0.531	0.547	10.750	12.500	4,500	3,500	0.375	6,300	4,410	0.406
12A	0.750	0.812	1.090	1.170	0.531	0.547	14.000	15.000	4,500	3,500	0.375	6,300	4,410	0.406
13	0.750	0.875	1.090	1.265	0.531	0.563	10.875	13.250	4,500	3,500	0.375	7,150	5,000	0.438
13A	0.750	0.875	1.090	1.265	0.531	0.563	14.250	15.500	4,500	3,500	0.375	7,150	5,000	0.438
14	0.812	0.875	1.170	1.265	0.547	0.563	11.750	14.000	6,300	4,410	0.406	7,150	5,000	0.438
14A	0.812	0.875	1.170	1.265	0.547	0.563	14.500	15.875	6,300	4,410	0.406	7,150	5,000	0.438
15	0.812	0.938	1.170	1.375	0.547	0.581	11.875	14.000	6,300	4,410	0.406	8,800	6,610	0.469
15A	0.812	0.938	1.170	1.375	0.547	0.581	15.000	16.250	6,300	4,410	0.406	8,800	6,610	0.469
16	0.875	0.938	1.265	1.375	0.563	0.581	12.000	14.000	7,150	5,000	0.438	8,800	6,160	0.469
16A	0.875	0.938	1.265	1.375	0.563	0.581	15.500	16.750	7,150	5,000	0.438	8,800	6,160	0.469
17	0.875	1.000	1.265	1.460	0.563	0.597	12.250	14.250	7,150	5,000	0.438	11,000	7,700	0.500
17A	0.875	1.000	1.265	1.460	0.563	0.597	16.000	17.125	7,150	5,000	0.438	11,000	7,700	0.500
18	0.938	1.000	1.375	1.460	0.581	0.597	13.188	15.750	8,800	6,160	0.469	11,000	7,700	0.500
19	0.938	1.062	1.375	1.600	0.581	0.670	13.750	16.500	8,800	6,160	0.469			0.531
20	1.000	1.062	1.460	1.600	0.597	0.670	14.000	16.375	11,000	7,700	0.500			0.531
21	1.000	1.125	1.460	1.680	0.597	0.670	14.250	16.375			0.500			0.563
22	1.062	1.125	1.600	1.680	0.670	0.670	15.063	18.000			0.531			0.563
23	1.062	1.250	1.600	1.930	0.670	0.860	15.875	18.500			0.531			0.625
24	1.125	1.312	1.680	2.000	0.670	0.920	17.000	19.000			0.563			0.656
25	1.250	1.312	1.930	2.000	0.860	0.920	17.375	19.500			0.625			0.656
26	1.250	1.375	1.930	2.050	0.860	0.920	18.250	20.500			0.625			0.688
27	1.250	1.438	1.930	2.100	0.860	0.920	19.000	21.378			0.625			0.719
28	1.312	1.500	2.000	2.240	0.920	1.020	20.000	22.000			0.656			0.751
29	1.438	1.500	2.100	2.240	0.920	1.020	21.000	23.000			0.719			0.750
30	1.438	1.625	2.100	2.440	0.920	1.020	22.000	24.000			0.719			0.813
31	1.500	1.688	2.240	2.530	1.020	1.110	22.500	24.500			0.750			0.844

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NOTES:

1. Dimensions are in inches.

2. The wrenches shall be of uniform quality. They shall be free from nicks, splits, cracks, laps, seams and scale. They shall have no imperfections that will adversely affect serviceability, reliability or safety.

3. Example of a military part number:

M8982/2 - 01

- 01 — Combination No. (small head=0.218, large head=0.281) see Table I.
- M8982/2 — Basic specification sheet number

4. For design purposes, this specification sheet takes precedence over procurement documents referenced herein.

Custodians:
Army - AV
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Navy - AS

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Navy - AS

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